

HAT-P-37b

R-2563

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_150705 · created 2026-08-26T23:20:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457208.713 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.7 +/- 1.58 [%]
Orbital Inclination (inc)	84.31 +/- 2.38
Airmass coefficient 1 (a1)	0.032 +/- 0.03
Airmass coefficient 2 (a2)	3.0 +/- 1.17
Scatter in the residuals of the lightcurve fit is	103.35 %
Best Comparison Star	None
Optimal Aperture	3.45
Optimal Annulus	12.2
Transit Duration (day)	0.06 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1394 (deviation 0.21σ)
Duration fit vs published	0.06 d vs 0.0983 d (deviation 0.47σ)
Mid-time O–C	2.5 min
Depth SNR	1.4
Residual scatter	103.35 %

Light curve

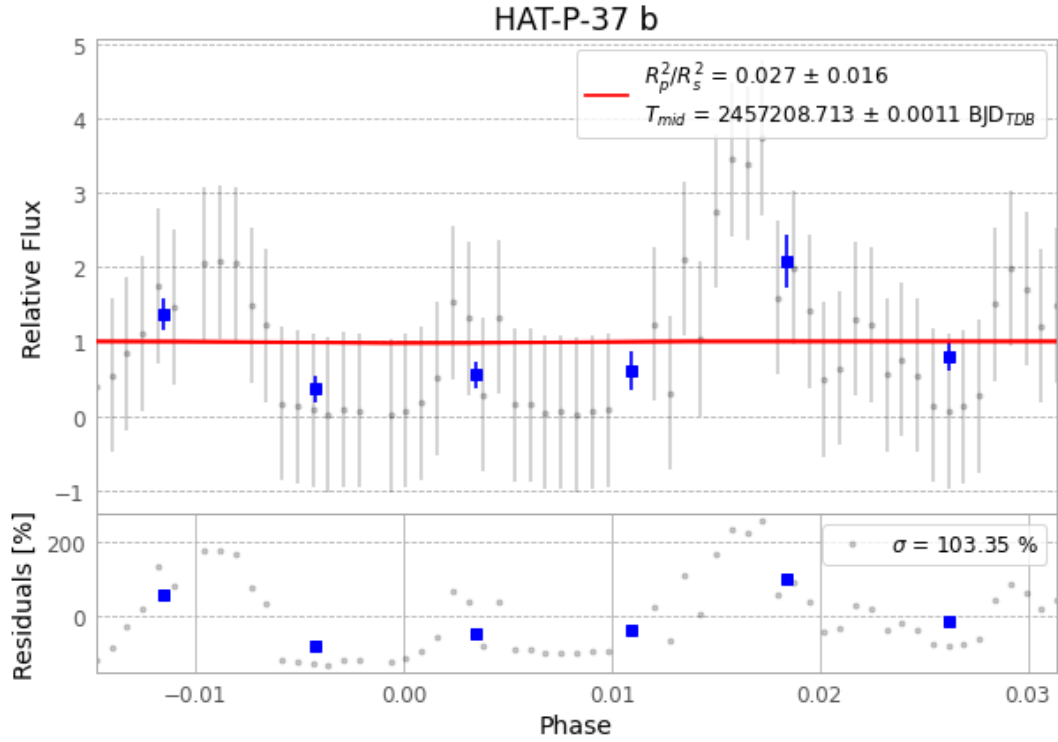


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-07-04.png (SHA-256 d94e4f1e827273c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [183, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f67adb6510822f9d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), HATP-37150705040312.FITS → HATP-37150705070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150705.log (bbefda80559e...), FinalLightCurve_HAT-P-37 b_2015-07-04.png (d94e4f1e8272...), FinalLightCurve_HAT-P-37 b_2015-07-04.csv (fc1784692653...)

Compute resources

Wall time 164.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:20Z.